

Literature Review in Sustainability – Greenwashing in E-commerce, User Interface Design Approach

MEC-E3007 – Product Sustainability

Aku Paloheimo 909282

Table of Contents

Introduction	3
User interfaces	3
Greenwashing	4
Why is it important?	4
Current state	5
Design patterns	6
Discussion	9
References	10

Introduction

In computer science and human computer interaction (HCI), sustainability is still structurally under-theorized and frequently overlooked. The environmental impact is not as tangible as in physical product development. The environmental impact of digital products, especially mass-market ecommerce applications, is, however, significant (García-Mireles et al.,2025). Both directly, through energy usage, device churn, digital waste, and indirectly through shaping consumption norms, increasing throughput velocity, increasing material demand (García-Mireles et al.,2025). Especially with the increased usage of artificial intelligence, the energy demand of the computer science industry has risen significantly (Li et al. 2025).

At the same time user interface design choices in mobile products increasingly are linked to environmental meaning. Green and minimalist visual styles are repeatedly culturally associated with eco-friendliness, care and “low impact”. For example, leftover food applications like ResQ and fiksuruoka.fi, as well as large corporate players such as Fortum and Neste, utilize visual patterns and interaction patterns that imply greenness, responsibility, circularity and “good impact” regardless of the actual verified sustainability actions of their operations.

This creates a design ethics problem: green user interfaces do not mean green impact. Additionally mobile ecommerce apps are now a primary vector where this symbolic greenness is performed and normalized on a global scale.

This literature review therefore maps existing academic work on greenwashing, dark patterns, sustainability in computer science and visual cues, to identify which specific interface design choices enable, amplify or legitimize greenwashing claims inside mobile e-commerce applications. The purpose is to recognize design patterns that systematically mislead and shift user trust and perception, so that future designers, researchers and regulators can better detect, name, and constrain them.

This literature review tries to find answers to the following research questions: “What design patterns are common with greenwashing in mobile e-commerce contexts?” and “Which design patterns shape customer perception of the sustainability of a brand?”

User Interfaces

In the context of this literary review user interfaces will be discussed at a higher level than other HCI reports. For instance, this literary review will not analyze human-centered design, that is an approach to systems design and development that aims to make interactive systems more usable by focusing on the use of the system by applying human factors, ergonomics, usability knowledge, and techniques in the development process (International Organization for Standardization, 2019). However, it is important to recognize, since modern user interface development depends a lot on human-centered design.

In that regard, it will be important to define user interfaces to analyze them deeper. A user interface describes all the components of an interactive system, that may be software or hardware, that provide information and controls for the user to accomplish their task with that interactive system (International Organization for Standardization, 2019).

Additionally, in this literary review we need to discuss user experience in the context of user interfaces. Since they determine the way users perceive and respond to the use or anticipated use of a product, system or service (International Organization for Standardization, 2019). Also to distinguish the parts of a system that make it usable

from what is not necessary we need to also define usability. Usability is the extent to which a system, product or service can be used by the specified user to achieve their goals with effectiveness, efficiency and satisfaction in a specified context of use (International Organization for Standardization, 2019).

Greenwashing

Greenwashing refers to communication strategies where organizations intentionally overstate, distort, or lie about environmental responsibility to generate positive perception in the eyes of the public without actually providing the claimed environmental impact (Botea-Muntean & Constantinescu, 2023; Sailer et al., 2022; Saleh et al., 2025). Often greenwashing is conducted via marketing by having vague claims, unverifiable claims, emotionally appealing symbolic cues, or exaggerated environmental benefits used to influence consumer decisions. Usually, to back up the sustainability claims, companies will accompany their statements with sustainability data (Botea-Muntean & Constantinescu, 2023; Sailer et al., 2022; Saleh et al., 2025). However, when the transparency on the data in the market is low greenwashing has an opportunity to be used as a tool for profit maxing (Saleh et al., 2025).

Greenwashing is happening increasingly (Sailer et al., 2022). This means there is value to companies that do it, but interestingly it often does not make sense, because there can be large financial losses (Botea-Muntean & Constantinescu, 2023; Saleh et al., 2025). On the contrary, when detected it can lead to the company to do more ecological actions to back up their previous unbacked sustainability claims (Botea-Muntean & Constantinescu, 2023).

As commerce and marketing shift from physical retail to the web, the primary medium of greenwashing is going to be user interfaces. The design of interfaces is used to promote corporate sustainability, and mobile apps provide a low entry point for this (Isensee et al., 2022). Color choice, typography, motion language, content ordering, labels, gamification, and persuasive technologies all can visually and interactionally create perceived pro-environmental behavior regardless of supply chain reality (Isensee et al., 2022).

Why is it important?

Greenwashing illegitimizes the companies and organisations that are doing actual work to better the world we live in. Users are very aware of their consumption, therefore design patterns in apps that they use can lead the user to accuse a company of greenwashing (Größbacher et al., 2025; Lawson et al., 2025). As a result, it makes the work that the real green companies do that much more difficult. Additionally, greenwashing is quite difficult to recognize as a consumer or a researcher (Sailer et al., 2022). But when recognized it can have true impact on the organizations eco-friendliness, by forcing the company to become more sustainable or lose face in the eyes of their customers (Botea-Muntean & Constantinescu, 2023).

As user interfaces are getting more complex and being used increasingly for marketing, it becomes important to also recognize the patterns that get used to mislead user, so called dark patterns (Barboza & Filho, 2019; de Jonge et al., 2025). That is why studies like this one are important to ease recognition and prevention of greenwashing. Similarly to dark patterns, greenwashing in user interfaces can distort consumer decision making (Größbacher et al., 2025; de Jonge et al., 2025). It can also reward the appearance of sustainability instead of actual impact and create

“performative greenness” as an acceptable design language. For human computer interaction and design research this is a design ethics problem and an epistemic problem. When the interface layer itself becomes an environment where truth is ambiguous, sustainability literacy degrades (Altammami et al., 2024).

Since mobile ecommerce apps are now one of the main infrastructures of consumption, understanding how user interface design patterns can implicitly greenwash is critical for regulation and responsible design standards. This is why a design pattern mapping is necessary, to make user interface patterns legible, nameable, and therefore governable.

Current state

E-commerce shows a growing normalization of greenwashing practices, driven by the rapid development of digital retail and increasing consumer interest in sustainability (Botea-Muntean & Constantinescu, 2023; Isensee et al., 2022; Sailer et al., 2022). Research shows that consumers frequently rely on visual cues, like color schemes, icons or product descriptions, when evaluating environmental claims in online environments while experts evaluate sustainability across the full product lifecycle, including manufacturing, logistics, disposal, and recycling (Altammami et al., 2024). This gap in sustainability literacy creates a fertile ground for companies to oversimplify or exaggerate ecological benefits through interface design choices that visually imply sustainability without substantiating it (Isensee et al., 2022).

At the same time empirical research shows that greenwashing strategies are widely employed across digital marketing channels including social media and online shopping (Sailer et al., 2022). For instance, small sustainable fashion enterprises have been found to use these tactics during sales such as Black Friday, unintentionally legitimizing deceptive patterns for less sustainable competitors and contributing to a decreased understanding of sustainability within e-commerce interfaces (Sailer et al., 2022).

Additionally, user interfaces play a key role in this relationship. Visual nudges, emotional cues, and persuasive interaction patterns can influence consumer decision making but they also risk being perceived as manipulative or misleading (Größbacher et al., 2025; Saleh et al., 2025; Szabo & Webster, 2021). Nudges in the context of user interfaces mean that the interface is trying to guide the user to act in a specific way (Isensee et al., 2022). Research on sustainable delivery nudges shows that while such cues can shift user behavior, they may simultaneously reduce user trust satisfaction and perceived autonomy (Größbacher et al., 2025; Saleh et al., 2025; Isensee et al., 2022). Especially among customers who are very sustainably aware or sensitive to persuasive design (Größbacher et al., 2025; Isensee et al., 2022). Additionally, since e-commerce is increasingly online, consumers are at the same time increasingly interested in the production process and the sustainability of the products that they purchase (Barboza & Filho, 2019). Also, it is shown that organizations increasingly integrate nudging and gamification into their digital applications, often blurring the line between encouragement and manipulation (Isensee et al., 2022). It is suggested that nudging will likely gain more relevance than gamification in future interface design (Isensee et al., 2022).

Finally, research on asymmetrical communication between parties that have different amounts of prior information, shows that sustainability marketing can significantly increase consumer trust, but greenwashing has an equally strong negative effect (Saleh et al., 2025). Trust is a critical predictor of purchasing behavior,

and consumers with higher environmental knowledge are particularly sensitive to misleading cues, making transparent, evidence-based claims essential for credible e-commerce applications (Saleh et al., 2025).

Overall, the current state of greenwashing in e-commerce reveals a tension between the persuasive power of interface design and the growing demand for authenticity. As digital retail continues to shape consumption norms, the user interface becomes a crucial tool in sustainability communication, where the organizations' sustainable actions are either communicated, distorted, or clarified. In this regard we can see the growing demand for sustainability from consumers and the increased responsibility companies are taking to ensure they are not trying to mislead consumers. Most likely if a company is not doing anything for sustainability the best bet is to try and not convince their customers otherwise.

Design patterns

As user interfaces play an increasingly important role in how sustainability is communicated, perceived, and potentially manipulated in e-commerce environments their design patterns become important tools for brand images. As discussed earlier, credibility and authenticity are essential for maintaining consumer trust (Bulmer et al., 2024). Interactivity of an application can increase the credibility of a company by making information easier to access and by creating a sense of responsiveness and connection between the user and the organization (Szabo & Webster, 2021).

At the same time, consumers' willingness to engage in sustainable behaviour is strongly linked to personal values (Barboza & Filho, 2019). When a digital interface supports value-based decision making, such as making environmental impacts visible and personally relevant, it increases the likelihood of pro-environmental choices (Barboza & Filho, 2019). This highlights the importance of design patterns that not only present sustainability information but also frame it in a way that aligns with user motivations. These kinds of design patterns can be seen in applications like *fiksuruoka.fi* and *ResQ Club* shown in Figure 1, where the whole business model, buying leftover food at a reduced price, is revolved around the consumers actions and their desire to be more sustainable.

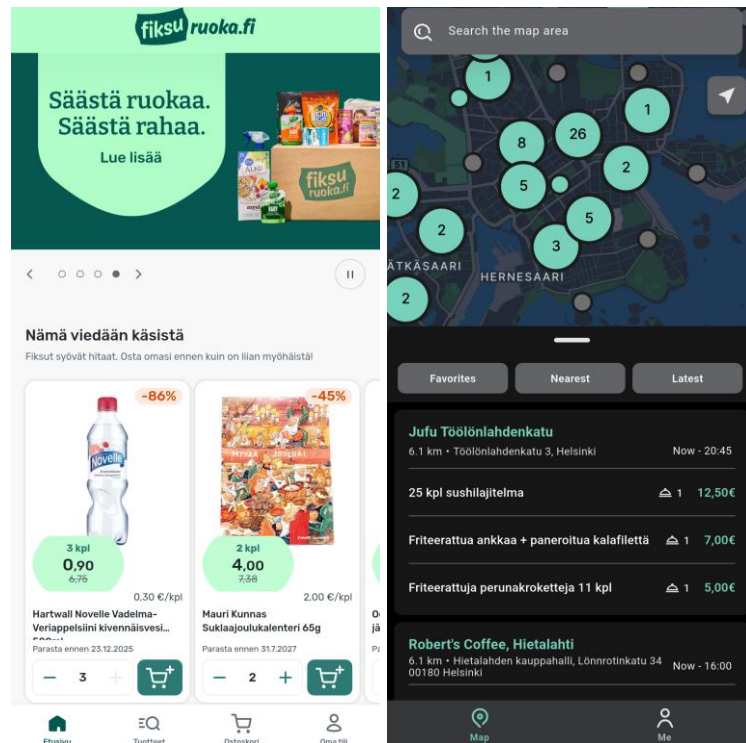


Figure 1. Fiksuruoka.fi and ResQ Club mobile applications user interfaces. The front-pages use green color to tell the user indirectly that the software is part of a sustainable organization.

A foundational approach to creating such patterns is human-centered design (HCD). HCD supports sustainability across the three pillars, economic, social, and environmental, by designing systems that are usable, accessible, and reusable (International Organization for Standardization, 2019). HCD ensures that sustainability related features, like green colours as in Figure 1, and eco-labels such as in Figure 2, are not merely decorative cues, but meaningful components that help users understand and reflect on their choices. Because well-designed interfaces are more likely to be used consistently and correctly, HCD indirectly promotes sustainable product lifecycles and responsible consumption (International Organization for Standardization, 2019).

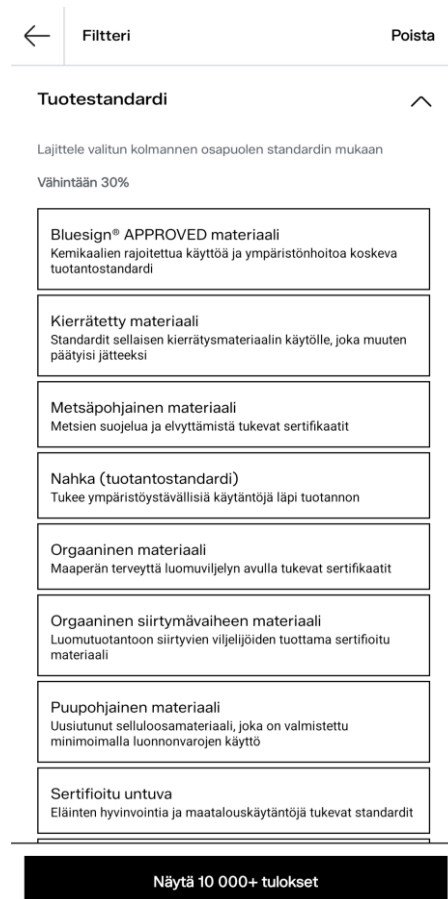


Figure 2. Zalando mobile application user interface. The user interface is from the filter selection and shows different sustainable labels that the user can choose to filter the results of their search.

However, design patterns can also be used to subtly guide users toward specific actions, such as “sustainable” delivery options or carbon-reduced products. Like how in Figure 2 sustainability labels are made more appealing. These nudges can promote positive behaviour, but as shown in prior research they may also be perceived as manipulative or greenwashed if they lack transparency or if emotional cues feel disproportionate to the actual environmental impact (Größbacher et al., 2025; Isensee et al., 2022). This presents a design dilemma: persuasive features can encourage sustainable action, but they must be designed carefully to avoid losing trust.

An approach to sustainable UI design involves enabling users to understand the impacts of their actions through feedback mechanisms (Barboza & Filho, 2019). Embedding environmental information directly into the product ecosystem for example, through smart meters that show real-time energy usage or cost savings (Barboza & Filho, 2019). In e-commerce, similar patterns might include carbon footprint meters, supply-chain transparency widgets, lifecycle indicators, or digital receipts that summarize environmental impacts. These patterns provide concrete, contextual data rather than symbolic greenness, thereby reducing the likelihood of greenwashing.

Taken together, the literature suggests that sustainable and trustworthy design patterns share three central characteristics. Transparency, users must be able to verify sustainability claims through accessible, meaningful information rather than symbolic

cues alone. Interactivity, Interfaces should enable exploration, comparison, and deeper understanding, increasing credibility. Value alignment, sustainable options must be framed in ways that resonate with personal motivations, without relying on manipulative emotional nudges.

Discussion

Based on the literature greenwashing has a negative effect on companies, and it has been proven that they should avoid doing it (Botea-Muntean & Constantinescu, 2023; Saleh et al., 2025). However, companies still find some sort of value in it and still partake in greenwashing (Sailer et al., 2022). So, greenwashing must have some monetary value to the company that does it, but it seems to be kind of a mystery. This shows us that greenwashing is here to stay and needs to be recognized.

The real issues lie within our society that allow the companies to get away with greenwashing, but all we can do as consumers is try our best to be critical of organizations and their sustainability claims. To aid this we can try to improve our green marketing literacy and try to recognize greenwashing when it is used against us.

Us consumers are increasingly interested in our consumption and increasingly aware of greenwashing. Which is a good direction in improving our green marketing literacy. However, when there is a force there is always a counter force. Companies are most likely trying to influence us consumers everywhere they can and gain our trust and influence our values.

As user interfaces are becoming increasingly popular in aligning companies' brand images the user interfaces at the same time become a platform for greenwashing. To recognize this in this literature review we have recognized some of the design patterns that try to gain our attention and try to influence us.

The first research question was "What design patterns are common with greenwashing in mobile e-commerce contexts?". To this there seemed to be no clear answer. For example, emotional nudges in applications are viewed as greenwashing due to their very emotional nature but were not used in a greenwashing attempt by the company. There are some common patterns like green color choices are often used in applications that have a green agenda, but a green color alone cannot be linked to greenwashing, since it could just be a part of the company color palette. The company might have their own view of what the color means. Additionally, sustainability labels are also often used, but they tend to be under heavy regulations and would be hard to use in greenwashing attempts.

The second research question was "Which design patterns shape customer perception of the sustainability of a brand?". For this there was literacy supporting some similar characteristics. Like transparency, accessibility, interactivity, and value alignment in applications positively affected customers' perceptions of companies. These findings give us some insight into how companies try to influence our purchasing decisions. In the information age companies are fighting for our attention and they want to be a part of our daily lives and to do that they will do their best to make themselves feel trustworthy.

REFERENCES

- Altammami, A., Dimitrova, V., & Pournaras, E. (2024). What you see, What you get? Mapping Inconsistencies of Sustainability Judgements among Experts and Consumers. *In Proceedings of the 2024 International Conference on Information Technology for Social Good (GoodIT '24)*. Association for Computing Machinery, 443–452.
doi:<https://doi.org/10.1145/3677525.3678695>
- Barboza, M. N. L., & Filho, E. J. M. A. (2019). Green Consumption Values in Mobile Apps. *Journal of International Consumer Marketing*, 31(1), 66–83.
doi:<https://doi.org/10.1080/08961530.2018.1490052>
- Botea-Muntean, D., & Constantinescu, R. (2023). GREENWASHING AND GREEN HUSHING. AN INTERNET OF THINGS APPROACH TO SUCH STRATEGIES. *Romanian Economic and Business Review*, 18(4), 7-20.
Retrieved from <https://www.proquest.com/scholarly-journals/greenwashing-green-hushing-internet-things/docview/3030877875/se-2>
- Bulmer, S., Palakshappa, N., Dodds, S., & Harper, S. (2024). Sustainability, brand authenticity and Instagram messaging. *Journal of Business Research* 175, 114547. doi:<https://doi.org/10.1016/j.jbusres.2024.114547>
- de Jonge, T., Schraffenberger, H., Geels, J., Hoepman, J. H., Simon, M. S., & Borgesius, F. Z. (2025). If Deceptive Patterns are the problem, are Fair Patterns the solution? *In Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*. Association for Computing Machinery, 3131–3137.
doi:<https://doi.org/10.1145/3715275.3732199>
- García-Mireles, G. A., Moraga, M. A., García, F., & Calero, C. (2025). Sustainability in the Field of Software Engineering: A Tertiary Study. *ACM Trans. Softw. Eng. Methodol.* Just Accepted (July 2025).
doi:<https://doi.org/10.1145/3747178>
- Größbacher, S., Ladner, S., Simhandl, J., Raffler, C., & Judmaier, P. (2025). Persuasive but Trustworthy? A Study of Sustainable Delivery Nudges in E-Commerce. *In Proceedings of the Mensch und Computer 2025 (MuC '25)*. Association for Computing Machinery, 684–688.
doi:<https://doi.org/10.1145/3743049.3748563>
- Isensee, C., Teuteberg, F., & Griesse, K. M. (2022). Exploring the use of mobile apps for fostering sustainability-oriented corporate culture: A qualitative analysis. *Sustainability*, 14(12), 7380. doi:<https://doi.org/10.3390/su14127380>
- International Organization for Standardization. (2019). Ergonomics of human-system interaction -- Part 210: Human-centred design for interactive systems (ISO Standard No. 9241-210). Retrieved from <https://online.sfs.fi/fi/index/tuotteet/SFS/CENISO/ID2/9/807069.html.stx#>

- Lawson, B. T., Coulentianos, M. J., & Mitchell, O. (2025). Can we trust green apps? mapping out 14 trustworthiness indicators. *Sustainability*, 17(14), 6444. doi:<https://doi.org/10.3390/su17146444>
- Li, P., Islam, M. J., & Ren, S. (2025) A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge. *ACM SIGMETRICS Performance Evaluation Review*, 53(2). 21-26. doi:<https://doi-org.libproxy.aalto.fi/10.1145/3764944.3764950>
- Sailer, A., Wilfing, H., & Straus, E. (2022). Greenwashing and bluewashing in black friday-related sustainable fashion marketing on instagram. *Sustainability*, 14(3), 1494. doi:<https://doi.org/10.3390/su14031494>
- Saleh, M. S. M., Li, S., & Huang, M. (2025). Green marketing and greenwashing effects on consumer purchase of electric cars: The moderating role of environmental knowledge. *International Journal of Customer Relationship Marketing and Management*, 16(1), 1-21. doi:<https://doi.org/10.4018/IJCRMM.367599>
- Szabo, S., & Webster, J. (2021). Perceived Greenwashing: The Effects of Green Marketing on Environmental and Product Perceptions. *Journal of Business Ethics*, 171(4), 719–739. Retrieved from <http://www.jstor.org/stable/45390484>